

When Do Uncertain Dates Change Retrieval? Exact Certificates for Temporal Evidence

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Abstract

Sources often date historical changes only to a month or year. Guessing exact dates can change which evidence reaches an answer model. We ask when that uncertainty matters to retrieval. Our compiler stores at most two replacement witnesses per claim. These witnesses identify evidence that could apply and evidence that must apply somewhere within the requested period. We give an exact test for whether every allowed timeline selects the same ranked context. When the test fails, we construct two valid timelines with different contexts. We also identify support decisions that every stabilizing date refinement must resolve. These operations take linear time, although general possible-context membership is NP-hard. A source-only TimeQA evaluation covers 2,348 validation questions. Among 375 parsed questions with modeled evidence in context, 306 have certified stable contexts. The remaining 69 expose consequential date ambiguity. The implementation preserves source spans, explicit departures, and unparsed evidence through answer generation.

1 Introduction

Historical questions often ask about a period, while sources give only approximate event dates. A source might report an appointment in September without identifying its day. Choosing the first or last day can change which evidence reaches a reader. Yet resolving every uncertain date wastes effort when those dates cannot change the selected context. We ask a practical question: *When does date uncertainty actually change retrieval?*

This problem arises before answer generation. Retrieval-augmented systems must select a few excerpts from longer histories (Lewis et al., 2020). Consider the presenters of a television programme during a transition month. The source names both presenters and dates their succession only to that

month. The previous presenter may contribute evidence for part of the requested period. A single guessed transition date conceals that uncertainty. A question covering the entire transition month can instead have a stable evidence set. The distinction depends on the question’s scope and the context budget.

Temporal memories already preserve histories and invalidate replaced facts. Graphiti applies direct contradiction decisions (Rasmussen et al., 2025; Graphiti contributors, 2026). NuggetIndex maintains atomic evidence, validity intervals, and lifecycle states (Zerhoubi et al., 2026). DeFuzzRAG and IA-RAG address fuzzy dates and interval retrieval (Chen et al., 2026; Wang et al., 2026). We study the decisions available before completing uncertain dates. The goal is an exact, inspectable interface between temporal evidence and ranked retrieval.

Each claim receives a range containing its unknown start date. A directed decision specifies which other claims can replace it. We compile these inputs into at most two source witnesses per claim. The first determines when support becomes impossible. The second determines when guaranteed support can fail. They answer every point or period query without enumerating timelines. *Possible support* means that a claim applies somewhere in the requested period under at least one allowed timeline. *Guaranteed support* means that it applies somewhere in that period under every allowed timeline.

The resulting masks also decide whether uncertainty changes retrieval. Fix a relevance ranking and a budget of k claims. If its first k possible claims are guaranteed, every timeline returns the same context. Otherwise, we identify a selected claim whose membership changes across two explicit timelines. These timelines can be replayed through the retrieval pipeline. They identify which dates need clarification.

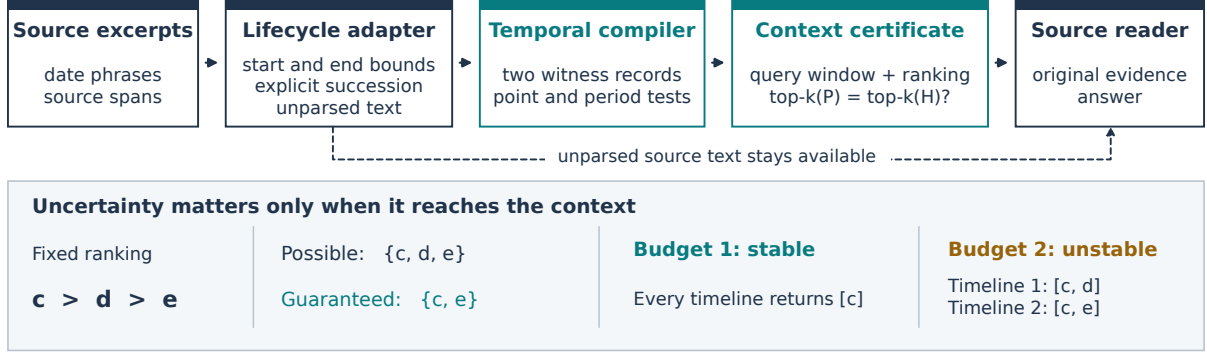


Figure 1: Source spans become dated claims with explicit replacement decisions. Two witnesses per claim support exact temporal tests. A fixed ranking then yields a stable context or two admissible timelines with different contexts. Unparsed source text follows a shared fallback policy.

Our contributions are threefold. First, we derive exact period-support formulas and a sharp two-witness representation. Compilation takes linear time in the claims and supplied replacement pairs. Second, we characterize context stability and construct witnesses when it fails. Date refinement preserves settled decisions, while each changed replacement pair affects only its target. We also separate tractable stability certification from NP-hard possible context membership. Third, we implement source-span extraction, explicit endpoints, matched date completions, and a local reader. Natural-source experiments measure extraction coverage, context ambiguity, and answer behavior under shared inputs. Complete-answer snapshot controls establish the costs of discarding alternatives before selection.

2 From Sources to Temporal Evidence

The source interface separates three meanings of time. An occurrence date says when a state began. A duration gives separate start and end dates. An annual observation says which values were reported for that year. Table 1 shows their different operations.

Source meaning	Representation
Started in September	An unknown start between that month’s first and last dates.
Served from 2001 to 2005	Separate uncertain start and end dates.
Listed in the 2005 snapshot	A dated observation retaining every listed answer.

Table 1: Date precision and state duration carry different information. Only an occurrence range bounds one unknown event date.

Occurrence precision. A day gives a point on the time axis. A month or year bounds an occurrence within that calendar period. Expressions such as *around 1713* require additional evidence before they supply finite bounds. We preserve the literal date expression and its source span. The query adapter applies the same convention to the visible question. A question about a whole year therefore remains a period query. Our predicate asks whether a claim applies *somewhere* within that period.

Replacement and coexistence. A directed replacement decision connects values of the same subject and role. For office succession, the organization and office define the slot; people supply its values. For affiliation, a new organization can coexist with an older affiliation. The Chicago Stadium passage in TimeQA, for example, gives overlapping tenancies for two teams (Chen et al., 2021). The source adapter records explicit succession separately from ordinary appointments or joins. All temporal policies receive identical pair decisions.

Explicit ends. A stated departure supplies an end event targeting its own claim. The compiler treats this hidden event as a direct replacement. Every permitted end must follow every permitted start: its lower bound exceeds the start’s upper bound. This reduction preserves gaps between tenures. It also handles a departure without a named successor. Appendix E proves equivalence to the corresponding half-open lifetime.

Evidence units. A paragraph can contain several states, so temporal selection operates on source excerpts. Each modeled claim preserves exact character offsets and its full supporting sentence. Uncovered text remains available through a shared

fallback policy. A loan’s planned expiry never becomes an observed departure. An incomplete temporal parse retains the complete source text. Readers receive original excerpts, their date bounds, and their support tags.

Support tags. Modeled claims are guaranteed, unresolved, or unsupported for the requested period. Unresolved claims have possible support without guaranteed support. Unmodeled excerpts carry a separate *unknown* tag. Their fixed fallback eligibility lets us isolate uncertainty in modeled dates. The context certificate concerns that declared temporal input and retrieval policy.

3 Dates as Constraints

Occurrence dates and replacement. Each claim c has text and an unknown occurrence date x_c . Its source bounds this date by two inclusive dates, $\ell_c \leq x_c \leq u_c$. Dates use a common real-valued time axis. These bounds describe uncertainty about an occurrence, rather than its duration. The allowed assignments form the product $\Omega = \prod_c [\ell_c, u_c]$. A fixed text gate $G(d, c)$ accepts a direct replacement when d occurs strictly after c . The gate need not be transitive. It excludes self-pairs: $G(c, c) = 0$. It must distinguish replacement from coexistence. The compiler preserves its decisions across date assignments.

For assignment x , claim c is active at date t when

$$A_x(c, t) = \mathbf{1}[x_c \leq t \wedge \nexists d : G(d, c), x_c < x_d \leq t]. \quad (1)$$

Equal dates do not establish replacement order. A question specifies an inclusive date window $Q = [a, b]$. Point questions set $a = b$. A claim supports the window when it is active somewhere inside it:

$$S_x(c, Q) = \mathbf{1}[\exists t \in Q : A_x(c, t) = 1]. \quad (2)$$

Possible support requires one allowed assignment with $S_x(c, Q) = 1$. Guaranteed support requires this condition in every allowed assignment. Write $P_Q(c)$ and $H_Q(c)$ for these two Boolean support decisions.

Two direct witnesses. One witness can force retirement despite date uncertainty. Another can permit retirement under a particular assignment. Their boundaries are

$$p_c = \min\{u_d : G(d, c), \ell_d > u_c\}, \quad (3)$$

$$r_c = \min\{\ell_d : G(d, c), u_d > \ell_c\}. \quad (4)$$

An empty minimum is $+\infty$. We store one witness for each finite boundary. These compiled certificates require a complete candidate scan. The compiler evaluates candidate gates and updates these two minima. It uses at most two witness records per claim. Two records are necessary in the worst case among direct witness subsets.

Theorem 1 (Exact window support). *For every claim and inclusive query window,*

$$P_Q(c) \iff \ell_c \leq b \wedge a < p_c, \quad (5)$$

$$H_Q(c) \iff u_c \leq b \wedge (a \leq \ell_c \vee a < r_c). \quad (6)$$

The first condition gives the smallest filter that never removes potentially supported evidence. The second gives the largest filter whose retained claims always have support. Each decision takes constant time after compilation. Appendix A gives the proofs.

Guaranteed support allows its supporting time to differ across timelines. For example, let $x_c \in [0, 10]$ and let a replacement start at time 5. Claim c always contributes somewhere in $[0, 10]$, either before the replacement or after its own later start. Yet no single point supports c across all timelines. A certainly active interval therefore cannot replace the window test.

Corollary 1 (Refinement and local influence). *Narrowing date bounds can only shrink possible support and expand guaranteed support. If two gates differ on h directed pairs, either support mask changes for at most h claims. With fixed rankings, their filtered top- k contexts differ in at most $2 \min(k, h)$ items.*

Each directed pair affects only its target claim’s certificates. When all date bounds are exact, both filters recover earliest direct replacement. The formulas therefore extend the same policy without choosing arbitrary dates.

4 Certifying the Reader’s Context

4.1 Compile once, query each claim in constant time

The compiler receives claims, occurrence bounds, and accepted directed replacement pairs. It initializes both boundaries to infinity. Each supplied pair updates at most two minima. Figure 2 gives the complete scan. A stable source identifier breaks ties between witnesses.

Let n count claims and m count accepted pairs. Compilation takes $O(n + m)$ time and stores $O(n)$

Input: claims and accepted pairs (d, c) .
Output: boundaries and their source witnesses.

1. Set $p_c = r_c = \infty$ for every claim.
2. For each accepted pair with $d \neq c$:
3. If $\ell_d > u_c$, minimize p_c using u_d .
4. If $u_d > \ell_c$, minimize r_c using ℓ_d .
5. Retain one supplying witness for each minimum.

Steps 3–4 execute inside the pair scan.

Figure 2: Linear compilation of the two support boundaries. The resulting records answer every query window.

certificate records. A support decision takes $O(1)$ time per claim. A complete mask takes $O(n)$ time. Pair discovery remains the source adapter’s responsibility. The reference complete scan makes $n(n - 1)$ gate calls. An index can replace this scan when it preserves every accepted pair.

4.2 An exact stability test

Let π_q be a complete relevance ranking for question q , including all tie breaks. For a set S , write $\text{Top}_k(S)$ for its first k ranked claims. The query’s possible and guaranteed sets are $P(Q)$ and $H(Q)$. A realized timeline x supplies a support set $S_x(Q)$. The reader’s selected claim context is

$$R_x(q) = \text{Top}_k(S_x(Q)). \quad (7)$$

Filtering preserves the ranking and occurs before the budget.

Theorem 2 (Exact context stability). *Every allowed timeline returns the same context exactly when*

$$\text{Top}_k(P(Q)) = \text{Top}_k(H(Q)). \quad (8)$$

Equivalently, every selected possible claim has guaranteed support.

Proof. Set $C = \text{Top}_k(P(Q))$. If $C \subseteq H(Q)$, every world retains C and selects the same context. Otherwise, choose $c \in C \setminus H(Q)$. Some world supports c , while another does not. At most $k - 1$ possible claims outrank c . Every world supporting c therefore selects it. The two worlds return different contexts. For $k = 0$, every context is empty. $\square$

The test uses two filtered rankings without enumerating timelines. Unknown excerpts can participate through the fixed fallback mask. Their inclusion remains constant while modeled dates vary.

Claim selection precedes source rendering and any word-budget truncation. We report rendered-context agreement separately.

Choose a stable context budget. The same scan finds the largest stable prefix. Let r be the position of the first unresolved claim in the possible-support ranking. Every budget $k < r$ returns a fixed context, while every $k \geq r$ permits a context change. If no unresolved claim exists, every budget is stable. This follows because the first $r - 1$ possible claims are guaranteed. The claim at position r supplies a counterexample for every larger budget. The system can therefore report a stable prefix alongside the unresolved evidence requiring date refinement. This decision preserves the supplied relevance order.

4.3 A source-grounded example

TimeQA’s Warburg Institute passage names three successive directors (Chen et al., 2021). Henri Frankfort started in 1949, Gertrud Bing in 1955, and Ernst Gombrich in 1959. A deterministic clause adapter recovers these three starts and their two direct succession links from the paragraph. The passage separately dates Bing’s earlier organizational membership to 1922. That membership date does not start her directorship.

Consider the illustrative period January 1954 through June 1955. Frankfort has guaranteed support somewhere within that period. Bing has possible support because her appointment might precede July. Gombrich has no support within the period. Rank Frankfort first and Bing second. A one-item context always contains Frankfort. A two-item context changes when Bing’s start moves from January to December 1955. Both assignments satisfy the source’s year-level precision. The same possible pool thus yields different stability decisions at different budgets. The source is natural; the query, ranking, and assignments illustrate the theorem.

4.4 Constructing and resolving an unstable context

Let $U = \text{Top}_k(P(Q)) \setminus H(Q)$. These claims have unresolved support and enter the possible context. Their status determines whether date uncertainty can change retrieval.

Theorem 3 (Constructive instability and refinement). *If $U \neq \emptyset$, we construct two complete date assignments with different contexts in $O(n)$ time.*

Here, n counts the event dates. Every date refinement that stabilizes the context makes each claim in U guaranteed or impossible.

Each $c \in U$ has fewer than k possible claims above it. Any timeline supporting c therefore selects it. We construct that timeline by assigning every other event before c or after a supporting query time. A second timeline places c after the query, or a direct witness by the query start. Date refinement can only remove possible claims. Thus a remaining possible pivot cannot leave the possible context through rank changes. It must become guaranteed. The assignments use only event bounds and the compact certificate. Full context replay takes $O(n + m)$ time with the accepted pair list. Appendix F gives the construction and proof.

The set U identifies support decisions that every stabilizing refinement must settle. New unresolved claims can enter after current pivots disappear. We recompute the frontier after each refinement.

4.5 A tractability boundary

A different question asks whether a given claim enters the context under *any* allowed timeline. Its answer depends jointly on the eligibility of higher-ranked claims.

Theorem 4 (Possible context membership). *When k is part of the input, deciding possible context membership is NP-hard.*

This holds even for a point query and a bipartite gate with one edge layer.

Proof. Take a Set Cover instance with elements E , sets $S_1, \dots, S_s$, and budget $1 \leq B \leq s$ (Karp, 1972). Each set becomes a claim v_j with start bounds $[1, 3]$. Create $B + 1$ copies of each element, all starting at time 0. Add a candidate c starting at time 0. A pair $v_j \rightarrow c$ targets every copy of each element in S_j . No other pairs are accepted. Rank every set claim and element copy above c . Use point query 2 and context budget $B + 1$.

For a cover of at most B sets, start its set claims at time 1. Start the others at time 3. Every element copy retires before the query. At most B higher-ranked set claims remain active, so c enters the context. Conversely, a context containing c has at most B active higher-ranked claims. Its active set claims must cover every element. An uncovered element would leave $B + 1$ active copies above c and exclude it. Thus these claims define a cover of size at most B . The construction is polynomial because $B \leq s$. $\square$

Appendix F also gives a reduction where every start precedes or equals the query time. Our stability test instead needs one unresolved claim inside the possible prefix. Its rank guarantees selection whenever it has support. This makes exact stability and its witnessing timelines tractable without computing every attainable context member.

4.6 What the witnesses explain

Each certificate stores a target’s bounds, both minima, and their supplying source identifiers. An exclusion caused by retirement points to its direct replacement witness. An exclusion caused by a late start points to the target’s own date bounds. This makes the temporal decision inspectable without reconstructing a cluster history.

The two boundaries answer different questions. For $c = [0, 2]$, take accepted witnesses $d = [1, 4]$ and $e = [3, 5]$. Witness d supplies $r_c = 1$ because it can follow an early start of c . Witness e supplies $p_c = 5$ because it must follow every start of c . Keeping only d incorrectly allows c at time 5. Keeping only e incorrectly guarantees c at time 2. Neither one-witness subset reproduces every query decision.

The same direct structure limits update propagation. Adding events whose earliest dates exceed the query end preserves that query’s earlier support decisions. Later ingestion can still correct a historical date when its event bounds reach the queried period. These properties depend on direct temporal support, rather than document arrival order.

5 Evaluation Protocol

We evaluate three outputs: exact support decisions, context certificates, and evidence supplied to an answer reader. All new corpus inputs come from original TimeQA text (Chen et al., 2021). The source adapter receives article titles, paragraph text, identifiers, and paragraph order. The query adapter receives the visible question. Answer strings and annotated spans enter scoring only. Published annotation dates construct the official-template questions; inference reads only the resulting visible question.

Natural-source split. The development source pool contains 32,332 paragraphs from 740 articles. A fixed hash selects 50 eligible articles and 176 questions for the pilot. We freeze the adapter after source audits and correctness fixes. Removing additional calibration articles leaves 644 validation

articles and 2,348 questions. This split precedes validation outcomes. The appendix records every exclusion and the original split. Both tracks search the complete development source pool.

Matched temporal controls. We build one BM25 ranking with $k_1 = 1.5$ and $b = 0.75$ (Robertson et al., 1995). Every policy shares the extracted claims, explicit ends, replacement pairs, source excerpts, and tie breaks. The context holds five excerpts and at most 768 source-text words. Words are separated by whitespace; title and date metadata are additional prompt text. Unmodeled text follows the same fallback policy. We compare no filtering, earliest-date completion, midpoint completion, latest-date completion, possible support, and guaranteed support. An interval control tests start and explicit-end overlap while ignoring cross-claim replacement pairs. A separate latest-year baseline reranks the same twenty lexical candidates by source-mentioned years. This comparator changes ranking; the other policies preserve it.

Measurements. Exact span recall measures annotated offsets fully contained in the shown excerpts. Complete string coverage requires every annotated answer string. A parent paragraph match alone does not earn source-span credit. We report certificates over parsed questions and over contexts containing modeled evidence. All-fallback contexts receive a separate denominator. For each unstable validation context, we save two full date assignments and replay their selected source identifiers. Paired confidence intervals use 2,000 article-cluster resamples with seed 20261006.

Answer reader. We run Qwen2.5-1.5B-Instruct with official Q4_K_M weights and a pinned CPU runtime (Qwen Team, 2024). Temperature is zero, with seed 1729 and a maximum of 96 generated tokens. The prompt requests all source-supported answers and cited excerpt numbers in JSON. A hash selects 24 pilot questions before inference. Identical question–context prompts share one generation across methods. A separate diagnostic covers every pilot question whose date completions select different contexts. We freeze JSON parsing before inspecting responses. Full request hashes, raw responses, timing, and parsing failures accompany every score.

Annual answer controls. We retain the complete TempLAMA source arrays (Dhingra et al., 2022).

Context diagnostic	Pilot	Validation
All questions	176	2,348
Parsed windows	173	2,296
Modeled context, parsed	32	375
Certified within this group	23	306
Unstable within this group	9	69
Parsed all-fallback context	141	1,921

Table 2: Exact context decisions under the fixed fallback policy. Modeled-context rows require at least one modeled claim in the possible top-five context.

The held-out split contains 25,752 questions across 2,993 keys. These controls distinguish preserving accepted alternatives from choosing transition dates. They compare date-first ranking, latest-value selection, latest-batch selection, and a matched Graphiti policy kernel. Appendix B gives all policies, source reconstruction, and unchanged baseline runs.

6 Results

6.1 Date uncertainty and context uncertainty differ

Table 2 separates parsed questions from contexts containing modeled evidence. The validation adapter produces 481 claims, of which 221 have explicit end dates. These claims occupy 1.46% of the 32,976 indexed source units. Among 375 parsed questions with modeled evidence in the possible context, 306 have an unchanged-context certificate. Thus 81.60% of these contexts need no finer modeled dates to fix their selected evidence. The other 69 return different contexts under the matched date completions.

The certificate directs attention to the evidence actually selected. With modeled evidence anywhere in the first twenty lexical candidates, 987 of 1,056 parsed questions have stable contexts. Lower-ranked uncertainty can therefore remain unresolved. All 1,921 parsed all-fallback contexts also remain fixed under the declared policy. We report them separately because their dates remain unmodeled.

The context budget controls exposure to uncertain dates. We retain the same 375-question group and vary only the number of selected source units. Table 3 reports the exact stable-prefix decisions. Stability falls from 92.27% at one item to 79.73% at twenty items. Most of the decrease occurs within the first five positions. The diagnostic lets a system distinguish a short stable context from

Budget k	1	3	5	10	20
Stable (%)	92.27	85.33	81.60	81.07	79.73

Table 3: Budget sensitivity on the fixed group of 375 parsed validation questions with modeled evidence at $k = 5$. The unit-level certificate precedes source-text truncation.

a larger context requiring refinement. It measures selection before source-text truncation.

6.2 Replaying an unstable selection

For each of the 69 unstable validation contexts, the constructive procedure selects an unresolved claim. One assignment supports that claim within the query; another excludes it throughout the query. The replay records include all event dates, source offsets, and selected identifiers. They connect the abstract certificate to concrete reader inputs. All 138 constructed worlds replay exactly and produce different rendered contexts within each pair. In 47 cases, the excluding world retires the selected claim by the query start. In 22 cases, it moves the claim’s start beyond the query end. Sixty-seven contexts have one unresolved selected claim; two have two. These are the support decisions that any stabilizing date refinement must settle.

A date boundary changes the evidence. Ben Gummer’s pilot question asks about May 2017. The source dates his Ipswich parliamentary tenure to 2010–2017. Earliest completion chooses January 1, 2017 and removes the tenure excerpt. Latest completion chooses December 31 and retains it. The vacated position receives an unrelated heading from the same article. The certificate traces this difference to the explicit end event. Refining its bounds entirely after the query start settles inclusion. An upper end bound at or before that start settles exclusion. Additional precision therefore needs to resolve this boundary; an exact departure day is unnecessary.

6.3 Ranking controls separate evidence quality from stability

Table 4 reports exact shown-span coverage under matched source units. Possible support and the interval control coincide in this run. The source adapter supplies explicit endpoints and no cross-claim replacement pairs. Guaranteed support gains 0.09 points of span recall over the interval control. Its paired interval is $[-0.09, 0.26]$ points.

Policy	Span recall	All strings
No temporal filter	38.49	37.56
Earliest completion	39.20	38.25
Midpoint completion	39.07	38.12
Latest completion	39.03	38.07
Interval control	39.07	38.12
Possible support	39.07	38.12
Guaranteed support	39.15	38.20
Latest-year ranking	53.51	52.51

Table 4: Validation evidence coverage on 2,348 questions. Values are percentages. Every policy shares the same source units and context budget. Latest-year ranking reorders the lexical candidates.

Latest-year reranking reaches 53.51% span recall, compared with 39.07% for the interval control. The paired difference is 14.44 points, with interval $[12.05, 16.86]$. This result identifies ranking as the main evidence-quality lever in this source adapter. The context theorem applies to either fixed ranking. The reported certificate uses the common lexical ranking so temporal policies share identical relevance decisions.

6.4 Answer generation under matched contexts

The fixed 24-question reader sample contains 192 method conditions and 52 unique generations. All outputs pass the frozen JSON parser. Each method achieves 29.17% exact answer-set accuracy and 31.94% strict answer-set F1. The temporal filters and date completions produce identical answer sets on this sample. Latest-year ranking changes fourteen answers while retaining the same aggregate strict score. The nine-question completion-sensitive diagnostic uses 28 unique generations, including six reused from the primary sample. Earliest and latest date completion change the answer set on four questions. Possible and guaranteed support also change four answer sets. Every method has 33.33% strict answer-set F1 on this selected diagnostic. These cases show that source-date choices can propagate through context selection into generated answers.

6.5 Complete answers require complete candidate sets

Annual controls expose a separate selection loss. On the full TempLAMA split, latest-value selection achieves 99.89% first-result accuracy but only 86.37% answer recall. Latest-batch selection retains the same first-result accuracy and raises an-

swer recall to 99.98%. Selecting only one value suppresses 13.63% of accepted annual answers before the final context budget. Its key-cluster confidence interval is [13.06, 14.21] points. The matched Graphiti and exact direct-first policies produce identical endpoints on all 45,866 source observations. These controls establish the exact-date reference and preserve the benchmark’s full answer scope.

6.6 Independent execution checks

Enumeration checks the compiler directly against its possible-world definition. All 253,050 support comparisons and 328,747 context-stability comparisons agree. Explicit-end verification adds 39,600 support comparisons and 29,700 stability comparisons. An independent check of the constructive extension validates 1,300 counterworld pairs across 12,925 enumerated worlds. A separate small-instance check verifies both directions of the Set Cover construction. These finite correctness checks accompany the proofs and remain separate from corpus measurements.

7 Related Work

Temporal memory and replacement. Graphiti closes earlier facts through direct contradictory evidence (Rasmussen et al., 2025; Graphiti contributors, 2026). MemStrata applies deterministic supersession to structured keys (Yadav, 2026). ERASE edits stored fact histories as new documents arrive (Li et al., 2025). NuggetIndex attaches source spans, validity intervals, and lifecycle states to atomic records (Zerhoubi et al., 2026). It distinguishes functional and multi-valued relations. These systems establish direct invalidation and cardinality-aware memory as existing mechanisms. Our contribution concerns exact compilation across uncertain occurrence dates under fixed direct decisions.

Temporal retrieval. Re³ combines fact grouping, conflict filtering, and temporal relevance (Cao et al., 2026). DeFuzzRAG adapts temporal granularity and filters inferred document scopes by query overlap (Chen et al., 2026). IA-RAG represents intervals and refines uncertain event dates (Wang et al., 2026). TA-RAG retrieves evidence across timelines (Lau et al., 2025). These methods motivate interval-aware evidence selection. We preserve all assignments within supplied occurrence bounds. The resulting compiler distinguishes possible sup-

port from guaranteed support within a window. The latter uses a different quantifier order from overlap with certainly active points.

Uncertain temporal data. Possible and certain temporal semantics have a substantial database foundation. Anselma et al. (2013) study valid-time indeterminacy and compact representations. Fan et al. (2012) study data currency without reliable timestamps. Amarilli et al. (2017b) analyze possible and certain answers over order-incomplete data. Our model restricts uncertainty to independent occurrence bounds with a fixed replacement predicate. Within this model, two direct witnesses answer all support queries. The witness bound and local influence result concern this specific compilation problem.

Uncertain ranked answers. Hua et al. (2008) study top- k membership probabilities under uncertain record membership. Amarilli et al. (2017a) study unknown ranking values constrained by a partial order. We keep relevance scores fixed and derive uncertain eligibility from direct temporal replacement. Within this model, context stability has a linear decision procedure and constructive witnesses. Possible context membership remains NP-hard when the budget is an input.

Temporal evaluation. TimeQA supplies natural passages and questions about dated histories (Chen et al., 2021). TempLAMA evaluates factual answers at yearly resolution (Dhingra et al., 2022). StreamingQA and FreshLLMs study changing information access (Liška et al., 2022; Vu et al., 2024). FRESCO distinguishes obsolete evidence from current but insufficient passages (An et al., 2026). Our protocol preserves complete answer arrays and original answer spans. It separates snapshot observations from occurrence bounds. This distinction prevents a benchmark’s date labels from silently becoming stronger temporal assumptions.

8 Conclusion

Date uncertainty becomes actionable when it changes selected evidence. Our certificates connect that decision to source witnesses, replayable timelines, and support claims that require refinement.

Limitations

The guarantees use independent occurrence bounds and fixed directed replacement decisions. Dependencies such as a known succession order can restrict the allowed timelines. The product model then gives conservative support bounds rather than an exact representation of those dependencies. Incorrect dates or pair decisions can also produce incorrect temporal conclusions. The compiler executes the supplied model; it does not establish the truth of extracted claims.

The query predicate asks whether evidence applies somewhere within a period. Questions requiring uninterrupted validity throughout a period need a different predicate. The compact certificates answer support and stability queries. Replaying complete contexts retains the accepted graph and uses additional audit state. The hardness result treats the context budget as an input. It does not establish hardness for a fixed budget of five.

The rule adapter models 1.46% of development source units and accepts no cross-claim replacement pairs. Its natural-data comparison therefore exercises uncertain starts and explicit ends. The two-witness cross-claim mechanism is verified through proofs, independent execution checks, and separate calibration cases. Unknown fallback text remains available but has no certified temporal support. Stable fallback-heavy contexts must not be interpreted as complete temporal understanding.

The new reader study uses a small quantized model and a 24-question pilot. Its completion-sensitive diagnostic is selected by context disagreement. Neither establishes gains for larger readers or broad answer-generation workloads. TimeQA anchors can be aliases, headings, or mentions outside the relevant period. Strict annotated-answer metrics differ from expert judgments of semantic correctness. Additional source entailment annotation would improve the empirical assessment.

The earlier TimeQA human and template tracks overlap. The TempLAMA annual labels can combine concurrent values with within-year succession. The matched Graphiti comparison isolates its temporal policy kernel under supplied candidates. It is not a full-system comparison with its semantic extraction model. The release preserves original source licenses, raw results, and the scope of each experiment.

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A Proofs for Uncertain Dates

All date assignments belong to the product Ω defined in Section 3. The replacement gate remains fixed across these assignments. The proofs use real dates with closed input bounds. Every witness differs from its target: $G(c, c) = 0$.

A.1 Point support

Lemma 1 (Exact point support). *For every date t ,*

$$P_{[t,t]}(c) \iff \ell_c \leq t < p_c, \quad (9)$$

$$H_{[t,t]}(c) \iff u_c \leq t \wedge (t \leq \ell_c \vee t < r_c). \quad (10)$$

Proof. If $t < \ell_c$, no assignment makes c active. If $t \geq p_c$, a witness satisfies $\ell_d > u_c$ and $u_d \leq t$. It occurs strictly after c and by t in every assignment. Thus possible support requires $\ell_c \leq t < p_c$.

Conversely, assume these inequalities. Choose $x_c = \min(u_c, t)$. For each accepted witness, choose $x_d \leq x_c$ when its bounds permit that choice. Otherwise, $\ell_d > x_c$. If $x_c = t$, this witness occurs after t . If $x_c = u_c < t$, the definition of p_c gives $u_d > t$. Choose the witness after t in this case. Independent date bounds permit all these choices simultaneously. No witness then retires c by t .

For certain support, every assignment has $x_c \leq t$ exactly when $u_c \leq t$. Assume this condition. A witness can retire c by t exactly when some assignment satisfies $x_c < x_d \leq t$. Choose $x_c = \ell_c$. A suitable witness date exists exactly when $t > \ell_c$, $\ell_d \leq t$, and $u_d > \ell_c$. The minimum eligible ℓ_d is r_c . Possible retirement therefore requires $t > \ell_c$ and $r_c \leq t$. Its complement gives the stated condition. $\square$

The equality term is necessary. Take $[\ell_c, u_c] = [0, 0]$ and $[\ell_d, u_d] = [0, 1]$ with $G(d, c)$ true. Claim c is certainly active at zero. A strictly later replacement cannot have occurred by zero. It is not certainly active at any later date. An ordinary half-open interval cannot represent this singleton. The implementation retains the exact comparisons from the lemma.

A.2 Window support

Proof of Theorem 1. Possible point support occupies $[\ell_c, p_c]$ by Lemma 1. This set intersects $[a, b]$ exactly when $\ell_c \leq b$ and $a < p_c$. The two existential quantifiers commute, proving Equation 5.

For guaranteed support, first suppose $u_c > b$. The assignment $x_c = u_c$ gives no support within the window. Thus $u_c \leq b$ is necessary. Now assume that inequality. Every assignment with $x_c \geq a$ supplies support at x_c itself. For an earlier occurrence, support fails exactly when a direct witness retires c by a . Such an assignment exists exactly when some witness satisfies $x_c < x_d \leq a$. The preceding lemma shows this is equivalent to $a > \ell_c$ and $r_c \leq a$. Taking its complement proves Equation 6. $\square$

For each world, support means existence of a suitable time. The suitable time can differ across worlds. Consequently, a window can have guaranteed support even when no point has certain support. This distinction remains after exact text matching.

A.3 Sharpness and refinement

The possible filter retains precisely the union of world-specific support sets. Any smaller filter omits a supported claim in some world. The guaranteed filter retains precisely their intersection. Any larger filter includes an unsupported claim in some world. These statements establish the claimed minimal and maximal filters. They concern each retained claim’s support, rather than one jointly optimal context.

Narrowing date bounds replaces Ω with a subset $\Omega' \subseteq \Omega$. Existential support can only disappear when its assignment set shrinks. Universal support can only appear. This proves refinement monotonicity without assumptions about gate transitivity. For correlated dates, a product approximation instead gives conservative bounds. It can enlarge possible support and reduce guaranteed support.

A.4 Local gate influence

Suppose gates G and H differ on directed pairs with target set U . Every claim outside U uses identical terms in both boundary minima. Its support status therefore remains unchanged for every query window. Thus either mask difference lies within U , where $|U| \leq h$.

Fix a complete candidate ranking and deterministic tie handling. Changing one eligibility bit alters at most one selected item and one replace-

ment. Apply this argument successively to the targets in U . The resulting context difference is at most $2|U|$. Each context also contains at most k items. The difference is therefore at most $2 \min(k, |U|) \leq 2 \min(k, h)$.

A directed gate edit differs from a symmetric matching edit. A symmetric edit can change both directions and affect two target claims. Overlapping date bounds do not define one fixed older endpoint. A date edit can also affect several target certificates. The local bound does not treat either edit as one directed decision.

A.5 Exact dates and compilation cost

When $\ell_c = u_c = x_c$ for every claim, both boundary minima select the earliest strictly later direct witness. The possible and guaranteed masks then coincide for every window. For a point query, they equal the exact-date earliest-witness policy.

The reference compiler evaluates all $n(n-1)$ candidate directed gates. The certificates require this scan or a proved complete candidate index. Two stored witnesses alone cannot verify that no omitted witness changes a boundary. If accepted pairs are already available, two minima require $O(n+m)$ time for m pairs. The output contains at most $2n$ witness records. Every point or window decision takes $O(1)$ time per claim. A full mask takes $O(n)$ time. A complete text index can reduce gate evaluations without changing these results. No subquadratic guarantee is asserted for candidate discovery.

The worked certificate in Section 4 proves worst-case witness minimality.

B Source and Evaluation Details

B.1 TempLAMA source reconstruction

We recover the complete official test release and a public alias mirror. Both contain the same 34,963 records and the same answer identifiers. The normalizer verifies record IDs, query text, first answers, and full answer sets. The source manifest stores byte hashes for both inputs. It preserves all 45,866 answer observations.

A source key combines the subject identifier and relation identifier. The revised evaluation reserves all 300 previously used keys for legacy development. A deterministic key split assigns 764 additional keys to development and 2,993 to test. No key appears in two partitions. The test contains 25,752 yearly records and 33,908 answer obser-

983 vations. All thresholds and baseline choices use
984 development records.

985 A year is a benchmark observation bucket. It is
986 not a publication timestamp or an independently
987 observed update date. Multiple answers can reflect
988 within-year succession or concurrent values. An
989 answer absent from a later label set is not indepen-
990 dently verified as retired. The source audit conse-
991 quently reports changes in label sets. It does not
992 convert these changes into new occurrence bounds.

993 Textual names do not always identify one source
994 entity. Six parsed slots merge 13 source enti-
995 ties. Fifty identical question-year groups cover 105
996 source rows. Sixty-one of those rows occur in the
997 held-out test partition. The evaluation keeps these
998 cases. It does not expose hidden entity identifiers
999 to the text parser.

1000 B.2 TimeQA provenance and positive labels

1001 The source repository is pinned to commit
1002 38b0598; the manifest records its full identifier.
1003 The release includes immutable download URLs,
1004 SHA256 hashes, and Git blob checks. The nor-
1005 malizer preserves each original paragraph and its
1006 character offsets. Repeated relation histories reuse
1007 the same paragraph identifier. The test corpus con-
1008 tains 33,493 distinct paragraphs.

1009 The test annotations contain 384 empty answers
1010 with zero-length spans. The human subset con-
1011 tains 159 such annotations. The positive retrieval
1012 tracks exclude these placeholders. They retain
1013 2,613 template-rendered questions and 830 original
1014 human questions. The human track is a subset of
1015 the larger test collection. It is not an independent
1016 second benchmark.

1017 Eight articles occur in both the published devel-
1018 opment and test collections. Removing them before
1019 development produces 2,651 nonempty develop-
1020 ment questions. The natural-passage controls have
1021 fixed rules and no fitted parameters. The source
1022 audit records all removals and overlapping article
1023 paths.

1024 A matched string span identifies the benchmark’s
1025 retrieval anchor. Some anchors appear in headings
1026 or in passages about another period. Aliases can
1027 also create several labels for one entity. The evalua-
1028 tion therefore names its metrics annotated passage
1029 hit and annotated span coverage. Entailment re-
1030 quires a separate source-support judgment.

B.3 Source-grounded examples 1031

1032 The Chicago Stadium paragraph gives two over-
1033 lapping tenancies. The Blackhawks’ stated period
1034 is 1929–1994. The Bulls’ stated period is 1967–
1035 1994. Both cover the question’s period, 1967–1988.
1036 This example supports coexistence directly from
1037 the passage.

1038 The News Quiz paragraph describes several
1039 hosts between 1977 and 1995. It includes Barry
1040 Norman, Barry Took, and Simon Hoggart. Barry
1041 Took returns after an intervening host. This ex-
1042 ample illustrates range coverage and recurrence.
1043 Its multiple answers do not establish simultaneous
1044 hosts.

1045 A Ruud Lubbers paragraph mixes month-level
1046 and day-level employment dates. The associated
1047 question uses year-level bounds. Other passages
1048 use expressions such as *before 1516* and *around*
1049 *1713*. These examples motivate preserving source
1050 precision. They do not supply numerical uncer-
1051 tainty widths by themselves. The release records
1052 their original text and exact offsets.

C Archived Results and Source Support 1053

1054 The supplement preserves previous generations,
1055 metrics, and source files. These records use a
1056 selected-answer protocol. Their answer metric
1057 checks normalized target containment. It does not
1058 require the complete answer set. The new source-
1059 only reader runs use a separate complete-answer
1060 protocol.

1061 The previous TempLAMA builder kept the first
1062 listed answer. Its 300 keys contain 447 changes
1063 in that selected answer. In 153 cases, the previous
1064 answer remains in the new full answer set. The
1065 revised scorer instead accepts every official answer
1066 identifier. The annual controls use this complete-
1067 source protocol.

1068 The archived extractive reader accesses an anno-
1069 tated value field. The archived historical evaluator
1070 also uses a gold anchor. Those outputs remain re-
1071 producible as historical records. They are separate
1072 from the source-only controls in the main body.

1073 The RealProse source audit identifies 17 excerpts
1074 without their assigned subject-role statement. It
1075 identifies 33 excerpts without the assigned start
1076 year. One excerpt gives a different start year. These
1077 findings concern the supplied passages, rather than
1078 the underlying facts. TimeQA provides the primary
1079 natural-text retrieval source in this revision.

D Reproduction

The release includes the paper sources, vector figures, compilers, corpus normalizers, predictions, and result cards. A verification entry point reruns the compiler checks and reconstructs the paper tables. Longer entry points rerun full-source and natural-passage retrieval. Input and prediction hashes identify the exact evaluated artifacts. The Graphiti policy adapter includes the inspected upstream code segments and their license. It isolates temporal invalidation under supplied candidates. It does not reproduce Graphiti’s semantic extraction model.

E Explicit Lifetimes and Fixed Fallbacks

Let claim c start at $x_c \in [\ell_c, u_c]$ and end at $z_c \in [v_c, w_c]$. Assume $u_c < v_c$ and independent start and end dates. Its explicit lifetime is $[x_c, z_c]$, additionally shortened by accepted direct replacements. Introduce a hidden event e_c with bounds $[v_c, w_c]$ and the pair $e_c \rightarrow c$. Every world has $x_c < z_c$. The hidden event retires c exactly when the explicit end occurs. It therefore preserves activity in every world. The support boundaries update to

$$p'_c = \min(p_c, w_c), \quad r'_c = \min(r_c, v_c).$$

The same support formulas apply, with at most two witness records per claim. Hidden end events never enter a reader’s context.

The strict separation condition is necessary for this independent-date reduction. For start bounds $[0, 2]$ and end bounds $[1, 3]$, the product permits start 2 and end 1. A valid lifetime excludes that assignment, but direct replacement ignores an earlier event. The adapter therefore leaves overlapping start–end bounds unmodeled. It also retains approximate dates without justified finite bounds as unmodeled text.

Let F contain fixed fallback items and S_x contain the modeled support set. The operational mask is $S_x \cup F$. Its union over worlds is $P \cup F$, and its intersection is $H \cup F$. The context test therefore compares $\text{Top}_k(P \cup F)$ with $\text{Top}_k(H \cup F)$. Fallback eligibility records a policy choice and never receives the guaranteed-support tag.

Additional source constraints can restrict the independent date rectangle. Possible support then remains an outer bound, while guaranteed support remains an inner bound. Equal endpoint contexts

still certify stability under the extra constraints. Different endpoint contexts need not establish instability within that smaller family. Counterworld exactness uses the stated independent-date model.

F Constructive Context Uncertainty

Fix a query $Q = [a, b]$, a complete strict ranking, and a budget k . Let P and H be the exact possible and guaranteed support sets. Let F contain any source items retained under a fixed fallback policy. Fallback items have unknown temporal support. They enter every operational retrieval mask. Define

$$U = \text{Top}_k(P \cup F) \setminus (H \cup F).$$

The exact context theorem makes U empty precisely when the context is stable.

Construct two witnessing timelines. Choose $c \in U$. Fewer than k items in $P \cup F$ outrank c . Every world supporting c must therefore select it. Set $t = \max(a, \ell_c)$ and $x_c = \min(u_c, t)$. Possible support gives $t \leq b$ and $t < p_c$. For every nonpivot event d , assign

$$x_d = \begin{cases} \ell_d, & \ell_d \leq x_c, \\ u_d, & \ell_d > x_c. \end{cases}$$

The first case places d no later than c . The second places d after t . Indeed, if $x_c = t$, then $u_d \geq \ell_d > t$. Otherwise, $x_c = u_c < t$ and $\ell_d > u_c$. The definition of p_c then gives $u_d \geq p_c > t$. This rule also makes valid choices for all events outside the incoming witness set. Those choices cannot affect whether c is active. The assignments respect all independent date intervals. Claim c is active at t , so this world selects it.

For a second world, first consider $u_c > b$. Set $x_c = u_c$ and every other date to its lower bound. The claim starts after the query and cannot enter its context. Otherwise, failed guaranteed support gives $a > \ell_c$ and $r_c \leq a$. Choose the stored witness d for r_c . Set $x_c = \ell_c$ and $x_d = \min(u_d, a)$. The witness conditions give $\ell_d \leq x_d \leq u_d$ and $\ell_c < x_d \leq a$. Thus d retires c by the query start. Set every remaining date to its lower bound. This world cannot select c . The two contexts differ.

Let n count modeled events and fixed fallback items. The construction scans the event bounds and pivot certificate in $O(n)$ time. It returns $O(n)$ date values and needs no retained pair list. The pivot’s rank condition already proves that these assignments yield different contexts. Optional replay

scans the accepted pairs and ranking in $O(n + m)$ time. Explicit end events enter the same construction but remain hidden from the context. Direct context replay retains the accepted pairs and uses $O(n + m)$ audit state. The compact $O(n)$ certificate output stores support boundaries, rather than the full graph.

Every current pivot must be resolved. Narrow the date intervals without emptying any interval. Keep all other inputs fixed. Let P' and H' denote the refined support sets. Then $P' \subseteq P$ and $H \subseteq H'$. Take $c \in U$ that remains in P' . It still has fewer than k possible items above it. Thus $c \in \text{Top}_k(P' \cup F)$. If the refined context is stable, the exact context theorem requires $c \in H' \cup F$. Since $c \notin F$, it must belong to H' . Therefore, each current pivot becomes guaranteed or impossible.

This condition is necessary but not sufficient. For example, rank two claims c, d with independent start bounds $[0, 2]$. Use no replacement pairs, point query 1, and budget 1. Initially, the frontier is $\{c\}$. Refining c to start 2 resolves that pivot but exposes frontier $\{d\}$. The result therefore gives no minimum number of date refinements.

Hardness of the full possible context envelope. We reduce Set Cover (Karp, 1972) to possible top- k membership. The context budget k is part of the decision input. Take a universe E , sets $S_1, \dots, S_s$, and budget $1 \leq K \leq s$. Create variable claim v_j with date interval $[1, 3]$ for every set. Create $K + 1$ copies of each element claim, all dated exactly 0. Create candidate c dated exactly 0. Accept $v_j \rightarrow e$ for every copy of each $e \in S_j$. There are no other accepted pairs. The gate is bipartite, with one edge layer from variables to element copies.

Rank all variables and element copies above c . Set query time 2 and budget $k = K + 1$. Candidate c has guaranteed support. A variable is active exactly when its date is at most 2. Every active variable retires its associated element copies by the query time. An inactive variable starts after the query and cannot retire any copy by that time.

Given a cover of size at most K , assign its variables date 1 and others date 3. Every element copy is retired. At most K higher-ranked variables remain active, so the context includes c . Conversely, suppose a world includes c . At most K higher-ranked claims are active. The variables dated at most 2 form a collection of at most K sets. If they miss an element, its $K + 1$ active copies exclude c . They must therefore cover E . The construction has

polynomial size and proves NP-hardness. It does not establish hardness for a fixed budget such as $k = 5$.

The result also holds when every start occurs by the query time. For this variant, use variable bounds $[0, 2]$, element dates 1, and candidate date 2. Add a clock at date 1, ranked below the candidate, with a pair targeting each variable. Use the same variable-to-element pairs and query time 2. Variables before 1 retire under the clock. Variables at or after 1 remain active, but only those after 1 retire element copies. A cover assigns its variables 2 and all others 0. Conversely, a selected candidate permits at most K active variables. Those after 1 must cover every element, or $K + 1$ active copies exclude the candidate. Variables exactly at 1 add cost without covering an element. This variant has an acyclic gate with two edge layers.

This hardness concerns arbitrary possible context membership. It does not contradict the linear exact stability test. An unstable possible context already supplies a pivot with fewer than k possible items above it. That rank condition makes its supporting world sufficient to demonstrate a context change.

G Source Pipeline and Run Provenance

Development and validation. Removing the eight development–test article overlaps leaves 2,651 eligible development questions. A hash selects 176 questions from 50 articles for calibration. The remaining initial partition contains 2,475 questions from 679 articles. Source inspection and succession calibration identify 35 further validation articles for exclusion. Removing their 127 questions produces the reported 2,348-question validation partition. The amendment precedes validation outcomes and preserves the original split. Gold-anchor-guided source inspections and source-only succession selection have separate exclusion records. Rule development also inspected development source text outside the fixed pilot. It did not read answer labels or retrieval outcomes as inference features. The source manifests distinguish calibration inspection from the later frozen validation run.

Source adapter. A year maps to January 1 through December 31 on the Gregorian ordinal axis. A month maps to its actual first and last dates. A fully stated day gives a singleton. These are occurrence bounds, not inferred durations. Rules

require an explicit state-start or tenure construction. They reject relative durations, future loan expiries, and unsupported person resolution. Temporary-state qualifiers remain in the fallback source when no complete supported lifetime is extracted. All source words remain in disjoint modeled or fallback units. The compiled development corpus contains 481 claims and 221 hidden end events. No accepted cross-claim pair occurs under the frozen rule adapter. A separate calibration pool contains thirty source-cue-selected succession paragraphs. A narrow deterministic clause adapter extracts five dated starts and three direct pairs from two articles. Model-assisted source audits check role scope, date attachment, and replacement direction for each record. The Warburg case replays the automatic adapter’s output rather than manually supplied claim identifiers. Its query period and ranking remain illustrative.

Two exploratory learned adapters each process the first six calibration paragraphs. The first returns 41 records, all missing required source quotes. A schema-constrained variant returns 46 records; 45 fail literal source grounding. The remaining record has an incorrect holder–date association and fails semantic review. Both planned thirty-paragraph batches stop after six completed requests because their interface fails source grounding. No learned claim or pair enters the reported validation experiment. The partial manifests preserve settings, output hashes, and the stop decisions. These calibration counts do not estimate held-out extraction precision.

Reader generation and scoring. The model revision is 91cad511; the manifest records its full hash and weight checksum. The runtime is llama.cpp release b11435 at commit 43fe9c6. The official model license is Apache-2.0; the runtime license is MIT. Requests run sequentially with eight CPU threads and prompt caching disabled. Prompts include the visible question, original source excerpts, date bounds, and temporal status. The JSON response requests an answer list and excerpt indices. The scorer accepts a complete object or a single JSON code fence. It rejects narrative repair, invalid field types, invalid indices, and truncated outputs. Failures receive empty answer lists and remain in the denominator. Answer normalization applies Unicode NFKC, case folding, and word-token extraction. It preserves articles and compares deduplicated complete answer

Source-label property	Count	Rate (95% CI)
Multiple answers	6,498/25,752	25.2 [24.2, 26.2]
Old first retained	1,456/3,980	36.6 [35.1, 38.0]
Any answer retained	6,050/6,961	86.9 [86.0, 87.8]

Table 5: Answer-set properties on held-out source keys. The last two rows condition on first-answer changes and complete-set changes, respectively. Rates and intervals are percentages.

Policy	H@1 ↑	R@5 ↑	S@5 ↓
BM25	64.64	80.84	45.89
BM25 + recent ties	84.04	90.10	40.50
Age weighting	99.81	99.98	24.07
Date-first ranking	99.81	99.98	0.00
Original predicate	99.37	99.77	0.77
Graphiti kernel	99.78	99.93	0.00
Query latest-value	99.89	86.37	0.00
Query latest-batch	99.89	99.98	0.00

Table 6: Annual source-label retrieval on 25,752 held-out questions. All values are percentages. H@1: first-result accuracy; R@5: answer recall; S@5: target-label stale exposure.

sets. Normalized exact answer sets define the main generated-answer metric. Token overlap is retained only as a secondary metric.

The primary reader subset has 24 questions from 21 articles. Deduplication reduces 192 method conditions to 52 unique generations. A context-sensitive diagnostic has nine questions and 72 method conditions. Six primary generations are reused; twenty-two additional prompts complete that diagnostic. The cohorts overlap on two questions, giving 31 distinct questions and 74 actual generations. Selection uses disagreement among retrieved contexts, never answer quality. This slice characterizes sensitivity and is not an independently selected accuracy benchmark.

Reproduction. The release stores exact prompts, request hashes, raw responses, context membership, source offsets, and scoring outputs. It includes pinned public download instructions for the model and runtime. The generation weights are not embedded in the source archive. A reproducibility fix sorts BM25 token accumulation and stops after selecting five accepted units. Every one of 20,192 saved method contexts remains identical after this change. The original runner and equivalence logs remain available.

Policy	Human		Templates	
	H@5	R@5	H@5	R@5
BM25	45.66	45.04	32.49	31.75
Latest year	54.70	54.02	51.86	50.89
Year envelope	49.16	48.33	34.44	33.67
Explicit range	44.10	43.53	32.11	31.39
Range support	44.58	44.00	38.50	37.63

Table 7: TimeQA paragraph retrieval percentages. H@5 measures annotated-passages coverage; R@5 measures annotated-span recall. The tracks contain 830 human questions and 2,613 template questions.

G.1 Complete annual controls

The annual split and complete-source labels match the archived second-pass experiment. No repeated model inference is implied by these retained results.

G.2 Natural paragraph controls

These earlier paragraph runs use the published test passages and positive annotations. Their human-question results were inspected before the new development experiment. The source-unit validation and paragraph test results have different retrieval units and denominators.